

Material Safety Data Sheet



KAY LIMESHIELD

(In accordance with Article 10 Paragraph 1, Industrial Safety and Health Act)

Section 1. Chemical product and company identification

- a. **Korean name** : KAY 라임실드
Product name English : KAY LIMESHIELD
b. **Recommended use of the chemical** : Delimer
Restrictions on use : Use only for the purpose on the product label.
c. **Manufacturer/Supplier/Distributor** : Kay Chemical Company
8300 Capital Drive
Greensboro NC 27409
Emergency telephone number : 00798-14-800-8016

Section 2. Hazards identification

- a. **Hazard classification** : SKIN CORROSION/IRRITATION - Category 2
b. **GHS label elements, including precautionary statements**
Symbol : 
Signal word : Warning
Hazard statements : Causes skin irritation.
Precautionary statements
Prevention : Wear protective gloves. Wash thoroughly after handling.
Response : IF ON SKIN: Take off contaminated clothing and wash before re-use. Wash with plenty of soap and water. If skin irritation occurs, seek medical advice/attention.
Storage : Not applicable.
Disposal : Not applicable.
c. **Other hazards which do not result in classification** : Handling and/or processing of this material may generate a dust which can cause mechanical irritation of the eyes, skin, nose and throat.

Section 3. Composition/information on ingredients

Code : 924980-02

Ingredient name	Synonym	CAS number	%
citric acid	anhydrous citric acid; citric acid, anhydrous; 2-hydroxy-1,2,3-propanetricarboxylic acid; beta-hydroxytricarballic acid; 1,2,3-propanetricarboxylic acid, 2-hydroxy-	77-92-9	65 - 70
sodium carbonate	carbonic acid, disodium salt; crystal carbonate; soda ash; disodium carbonate; trona	497-19-8	25 - 30

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

Occupational exposure limits, if available, are listed in section 8.

Section 4. First aid measures

- a. **Eye contact** : In case of contact, immediately flush eyes with plenty of water. Remove contact lenses and flush again. Get medical attention if irritation persists.
- b. **Skin contact** : In case of contact, immediately flush skin with plenty of water. Wash clothing before reuse.
- c. **Inhalation** : If inhaled, remove to fresh air.
- d. **Ingestion** : Do not induce vomiting. Never give anything by mouth to an unconscious person. If irritation persists, get medical attention.

e. **Most important symptoms/effects, acute and delayed**

Potential acute health effects

- Inhalation** : Exposure to airborne concentrations above statutory or recommended exposure limits may cause irritation of the nose, throat and lungs.
- Ingestion** : Irritating to mouth, throat and stomach.
- Skin contact** : Causes skin irritation.
- Eye contact** : Causes serious eye irritation.

Over-exposure signs/symptoms

- Inhalation** : Adverse symptoms may include the following:
respiratory tract irritation
coughing
- Ingestion** : No specific data.
- Skin** : Adverse symptoms may include the following:
irritation
redness
- Eyes** : Adverse symptoms may include the following:
pain or irritation
watering
redness

f. **Indication of immediate medical attention and special treatment needed, if necessary**

- Specific treatments** : Not available.
- Notes to physician** : No specific treatment. Treat symptomatically. Contact poison treatment specialist immediately if large quantities have been ingested or inhaled.
- Protection of first-aiders** : No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation.

See toxicological information (section 11)

Section 5. Fire-fighting measures

a. **Extinguishing media**

- Suitable** : Use an extinguishing agent suitable for the surrounding fire.
- Not suitable** : None known.

b. **Specific hazards arising from the chemical** : No specific fire or explosion hazard.

- Hazardous thermal decomposition products** : Decomposition products may include the following materials:
carbon dioxide
carbon monoxide
metal oxide/oxides

c. **Special protective equipment for fire-fighters** : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

- Special precautions for fire-fighters** : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training.

- Remark** : Not available.

Section 6. Accidental release measures

- a. **Personal precautions, protective equipment and emergency procedures** : Do not allow to enter drains or watercourses. Put on appropriate personal protective equipment (see section 8). Provide adequate ventilation. Avoid breathing dust.
- b. **Environmental precautions** : Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).
- c. **Methods and materials for containment and cleaning up**
 - Small spill** : Move containers from spill area. Vacuum or sweep up material and place in a designated, labeled waste container. Dispose of via a licensed waste disposal contractor.
 - Large spill** : Move containers from spill area. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Vacuum or sweep up material and place in a designated, labeled waste container. Avoid creating dusty conditions and prevent wind dispersal. Dispose of via a licensed waste disposal contractor. Note: see section 1 for emergency contact information and section 13 for waste disposal.

Section 7. Handling and storage

- a. **Precautions for safe handling** : Put on appropriate personal protective equipment (see section 8). Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Workers should wash hands and face before eating, drinking and smoking. Do not ingest. Avoid contact with eyes, skin and clothing. Avoid breathing dust. Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use. Empty containers retain product residue and can be hazardous. Do not reuse container.
- b. **Conditions for safe storage, including any incompatibilities** : Store between the following temperatures: 0 to 50°C (32 to 122°F). Store in accordance with local regulations. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see section 10) and food and drink. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabeled containers. Use appropriate containment to avoid environmental contamination.

Section 8. Exposure controls/personal protection

a. Control parameters

Ingredient name	Exposure limits
None.	

- Recommended monitoring procedures** : If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment.
- b. **Appropriate engineering controls** : Use only with adequate ventilation. If user operations generate dust, fumes, gas, vapor or mist, use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits.
- Environmental exposure controls** : Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.
- c. **Personal protective equipment**
 - Respiratory protection** : A respirator is not needed under normal and intended conditions of product use.
 - Eye protection** : No protective equipment is needed under normal use conditions.

Section 8. Exposure controls/personal protection

- Hand protection** : No protective equipment is needed under normal use conditions.
- Skin protection** : No protective equipment is needed under normal use conditions.
- Hygiene measures** : Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period.
Appropriate techniques should be used to remove potentially contaminated clothing.
Wash contaminated clothing before reusing.

Section 9. Physical and chemical properties

- a. **Appearance**
- Physical state** : Solid. [Powder.]
- Color** : Green. [Light]
- b. **Odor** : Odorless.
- c. **Odor threshold** : Not available.
- d. **pH** : 4 to 4.6 [Conc. (% w/w): 1%]
- e. **Melting/freezing point** : Not available.
- f. **Boiling point/boiling range** : Not available.
- g. **Flash point** : > 100°C
- h. **Evaporation rate** : Not available.
- i. **Flammability (solid, gas)** : Not available.
- j. **Lower and upper explosive (flammable) limits** : Not available.
- k. **Vapor pressure** : Not available.
- l. **Solubility** : Easily soluble in the following materials: cold water and hot water.
- m. **Vapor density** : Not available.
- n. **Relative density** : 1.37 to 1.377
- o. **Octanol/water partition coefficient** : Not available.
- p. **Auto-ignition temperature** : Not available.
- q. **Decomposition temperature** : Not available.
- r. **Viscosity** : Not available.
- s. **Molecular weight** : Not applicable.

Section 10. Stability and reactivity

- a. **Chemical stability** : The product is stable.
- b. **Possibility of hazardous reactions** : Under normal conditions of storage and use, hazardous reactions will not occur.
- c. **Conditions to avoid** : No specific data.
- d. **Incompatible materials** : No specific data.
- e. **Hazardous decomposition products** : Under normal conditions of storage and use, hazardous decomposition products should not be produced.

Section 11. Toxicological information

a. Information on the likely routes of exposure

- Respiratory** : Exposure to airborne concentrations above statutory or recommended exposure limits may cause irritation of the nose, throat and lungs.
- Oral** : Irritating to mouth, throat and stomach.
- Skin** : Causes skin irritation.
- Eyes** : Causes serious eye irritation.

b. Delayed and immediate effects and also chronic effects from short and long term exposure

Acute toxicity

Product/ingredient name	Result	Species	Dose	Exposure
citric acid	LD50 Oral	Rat	3 g/kg	-
	LD50 Oral	Mouse	7280 mg/kg	-
	LD50 Oral	Mouse	5040 mg/kg	-
	LDLo Oral	Rabbit	7 g/kg	-
sodium carbonate	LD50 Dermal	Rabbit	>2000 mg/kg	-
	LD50 Oral	Mouse	6600 mg/kg	-
	LD50 Oral	Rat	4090 mg/kg	-
	LC50 Inhalation Vapor	Rat	2300 mg/m3	2 hours

Irritation/Corrosion

Product/ingredient name	Result	Species	Score	Exposure	Observation
Not available.					

Sensitization

Product/ingredient name	Route of exposure	Species	Result
Not available.			

Potential chronic health effects

- General** : No known significant effects or critical hazards.
- Inhalation** : Repeated or prolonged inhalation of dust may lead to chronic respiratory irritation.
- Ingestion** : No known significant effects or critical hazards.
- Skin contact** : No known significant effects or critical hazards.
- Eye contact** : No known significant effects or critical hazards.
- Carcinogenicity** : No known significant effects or critical hazards.
- Mutagenicity** : No known significant effects or critical hazards.
- Teratogenicity** : No known significant effects or critical hazards.
- Developmental effects** : No known significant effects or critical hazards.
- Fertility effects** : No known significant effects or critical hazards.

Chronic toxicity

Product/ingredient name	Result	Species	Dose	Exposure
Not available.				

Carcinogenicity

Product/ingredient name	Result	Species	Dose	Exposure
Not available.				

Mutagenicity

Product/ingredient name	Test	Experiment	Result
Not available.			

Teratogenicity

Product/ingredient name	Result	Species	Dose	Exposure
Not available.				

Reproductive toxicity

Section 11. Toxicological information

Product/ingredient name	Maternal toxicity	Fertility	Development toxin	Species	Dose	Exposure
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Not available.

c. ATE value

Name	Route	Result
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Other information : Not available.

Section 12. Ecological information

a. Aquatic and terrestrial toxicity

Environmental effects : No known significant effects or critical hazards.

Product/ingredient name	Test	Result	Species	Exposure
sodium carbonate	-	Acute EC50 347 mg/L	Daphnia	48 hours
	-	Acute EC50 199.82 to 298.9 mg/L Fresh water	Daphnia - Water flea - Neonate - 24 hours	48 hours
	-	Acute LC50 740 mg/L	Fish	96 hours
	-	Acute LC50 <850 mg/l	Fish	96 hours
	-	Acute LC50 320 mg/l	Fish	96 hours
	-	Acute LC50 300 mg/l	Fish	96 hours
	-	Acute LC50 1640000 to 2030000 ug/L Fresh water	Daphnia - Water flea - 24 hours	48 hours
	-	Acute LC50 1020000 to 1170000 ug/L Fresh water	Daphnia - Water flea - 24 hours	48 hours
	-	Acute LC50 <850000 ug/L Fresh water	Fish - Fathead minnow - 1 to 7 days	96 hours
	-	Acute LC50 740000 ug/L Fresh water	Fish - Western mosquitofish - Adult	96 hours
	-	Acute LC50 565000 ug/L Fresh water	Daphnia - Water flea	48 hours
	-	Acute LC50 320000 ug/L Fresh water	Fish - Bluegill	96 hours
	-	Acute LC50 300000 ug/L Fresh water	Fish - Bluegill - 3.88 cm - 0.96 g	96 hours
	-	Acute LC50 265000 ug/L Fresh water	Daphnia - Water flea	48 hours

b. Persistence/degradability

Product/ingredient name	Test	Result	Dose	Inoculum
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Not available.

Section 12. Ecological information

Product/ingredient name

Not available.

Aquatic half-life

Photolysis

Biodegradability

c. Bioaccumulative potential

Product/ingredient name

Not available.

LogP_{ow}

BCF

Potential

d. **Soil mobility** : Not available.

e. **Other adverse effects** : No known significant effects or critical hazards.

Section 13. Disposal considerations

- a. **Disposal methods** : The generation of waste should be avoided or minimized wherever possible. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements.
- b. **Disposal precautions** : Empty containers or liners may retain some product residues. This material and its container must be disposed of in a safe way. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

Section 14. Transport information

IMDG Class

- a. **UN number** : Not regulated.
- b. **Proper shipping name** : -
- c. **Classes** : -
- d. **Packing group** : -
- e. **Marine pollutant** : Not a pollutant.
- f. **Additional information** : -
- Label** :

Section 15. Regulatory information

a. Regulation according to ISHA

ISHA Article 37 : None of the components are listed.

ISHA Article 38 : None of the components are listed.

b. Regulation according to TCCA

TCCA Toxic chemicals : Not applicable

TCCA Observational chemicals : None of the components are listed.

TCCA Article 11 (Banned) : None of the components are listed.

TCCA Article 11 (Restricted) : None of the components are listed.

TCCA Article 14 (TRI) : None of the components are listed.

Korea inventory (KECI) : All components are listed or exempted.

c. **Dangerous Materials Control Act** : Not available.

d. **Wastes regulation** : Dispose of contents and container in accordance with all local, regional, national and international regulations.

e. Regulation according to other foreign laws

Europe inventory : All components are listed or exempted.

Section 15. Regulatory information

United States inventory (TSCA 8b) : All components are listed or exempted.

Japan inventory (ENCS) : All components are listed or exempted.

Japan inventory (ISHL) : All components are listed or exempted.

Section 16. Other information

- a. **References** : Not available.
- b. **Date of issue** : 4/1/2010.
- c. **Date of previous issue/Version** : No previous validation. / 1

☑ Indicates information that has changed from previously issued version.

Notice to reader

The above information is believed to be correct with respect to the formula used to manufacture the product in the country of origin. As data, standards, and regulations change, and conditions of use and handling are beyond our control, NO WARRANTY, EXPRESS OR IMPLIED, IS MADE AS TO THE COMPLETENESS OR CONTINUING ACCURACY OF THIS INFORMATION.